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General Remarks



Fundamental Concepts of Probability



Probabilities, Distributions, etc.



Bayesian Networks: Semantics



Inference in Bayesian Networks



Bayesian Parameter Learning



Structure Learning



Hidden Markov Models (HMMs)



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Started on Tuesday, 2 February 2021, 9:41 AM

State Finished

Completed on Tuesday, 2 February 2021, 11:00 AM

Time taken 1 hour 19 mins

Marks 36.00/56.00

Grade 6.43 out of 10.00 (64%)

Information

Flag question

- If asked for a single number, enter the number only, and omit the calculation or reasoning.
- If necessary, use a point to separate decimal places, e.g., 3.14. If needed, round to two decimal places.
- Wrongly formatted answers will be marked with a red triangle in the quiz navigation on the left (after clicking "Next page") and should be corrected before final submission.
- If anything is unclear, post in the Zoom chat, do not use the "Ask for help" button. We will either answer in the chat or ask you to come back to the main room (using the "Leave Breakout Room" link -- make sure not to leave the full meeting).
- Good luck!

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Ungraded toy exam ►

$$P(P, S = false) / P(S = false) = 0.2 / 0.3$$

Information

Flag question

Fundamental Concepts of Probability

Some questions about probabilities:

Consider a simplistic world of university exams, defined by the following full joint distribution over two random Boolean variables *Studied* (S) and *Pass* (P) $\in \{true, false\}$.

	$S = true$	$S = false$
$P = true$	0.5	0.1
$P = false$	0.2	0.2

Calculate the following things from this:

Important: You may round probabilities to two decimal places, e.g., $\frac{1}{3} \approx 0.33$ and $\frac{2}{3} \approx 0.66$.

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

$$P(S = false)$$

Answer:



$$0.1 + 0.2$$

The correct answer is



Question 2

Correct

Mark 1.00 out of 1.00

Flag question

$$\max P(P \mid S = \text{false})$$

Answer:

$$P(P=\text{true}, S=\text{false})/P(S=\text{false}) = 0.1/0.3 = 0.33$$

$$P(P=\text{false}, S=\text{false})/P(S=\text{false}) = 0.2/0.3 = 0.66$$

second one bcz it's max

The correct answer is:

Question 3

Correct

Mark 1.00 out of 1.00

Flag question

$$\arg \max_x P(P = \text{true}, S = x)$$

Select one:



$$P(P=\text{true}, S=\text{false})/P(S=\text{false}) = 0.1/0.3 = 0.33$$

$$P(P=\text{true}, S=\text{true})/P(S=\text{false}) = 0.5/0.3 = 1.66$$

second one bcz it's max

The correct answer is: ☐

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

$$P(P = \text{false})$$

Answer:

Question 5

Correct

Mark 1.00 out of 1.00

Flag question

$$\arg \max_x P(P = x \mid S = \text{false})$$

Select one:



The correct answer is: ☐

Question 6

Correct

Mark 2.00 out of 2.00

Flag question

$$P(S \mid P = \text{false})$$

	$S = \text{true}$	$S = \text{false}$
$P = \text{false}$	0.5	0.2

$$P(S=\text{true}, P=\text{false})/P(P=\text{false}) = 0.2/0.4 = 0.5$$

$$P(S=\text{false}, P=\text{false})/P(P=\text{false}) = 0.2/0.4 = 0.5$$

Question 7

Correct

Mark 2.00 out of 2.00

Flag question

$$P(P \mid S)$$

	$S = \text{true}$	$S = \text{false}$
$P = \text{true}$	0.5	0.1
$P = \text{false}$	0.2	0.2

$$= 0.5/(0.5+0.2) = 0.71$$

$$= 0.1/0.3 = 0.33$$

$$= 0.2/(0.5+0.2) = 0.28$$

$$= 0.2/0.3 = 0.66$$

Information
🚩 Flag question

Probabilities, Distributions, etc.

Which of the following statements are true?

Question 8
Incorrect
Mark 0.00 out of 1.00
🚩 Flag question

A likelihood function cannot take values greater than 1.0.

Select one:



The correct answer is



Question 9
Incorrect
Mark 0.00 out of 1.00
🚩 Flag question

A joint distribution $P(X_1, \dots, X_n)$ over several discrete variables cannot have fewer zeroes than each individual distribution $P(X_i)$.

Select one:



The correct answer is



Question 10
Correct
Mark 1.00 out of 1.00
🚩 Flag question

In a conditional distribution $P(X | y)$, all values $P(x | y)$ must be smaller than the corresponding $P(x)$ in the unconditional distribution $P(X)$.

Select one:



The correct answer is



Question 11
Correct
Mark 1.00 out of 1.00
🚩 Flag question

A continuous density function cannot take values greater than 1.0.

Select one:



The correct answer is



Question 12
Correct
Mark 1.00 out of 1.00
🚩 Flag question

In a joint distribution over variables X and Y , a probability $P(x, y)$ can never be larger than the corresponding $P(x)$.

Select one:



Question 13

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

If we normalize a likelihood function, we get a valid continuous density function.

Select one:



The correct answer is



Question 14

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

If X and Y are independent, then $P(X, Y) = P(X)$.

Select one:



The correct answer is

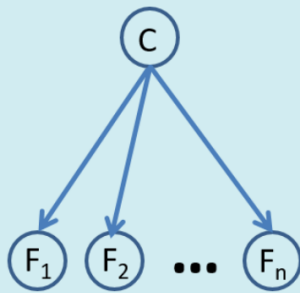


Information

🚩 Flag question

Bayesian Networks: Semantics

Consider the following Bayesian Network:



Information

🚩 Flag question

Which of the following expressions are correct statements about the full joint distribution represented by this network?

Question 15

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

$$P(C, F_1, \dots, F_n) = P(C) \times \prod_i P(F_i | C)$$

Select one:

☒ True☒ FalseThe correct answer is ☒**Question 16**

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

$$P(C, F_1, \dots, F_n) = P(C) \times P(F_1, \dots, F_n | C)$$

Select one:

☒ True☒ FalseThe correct answer is ☒**Question 17**

Correct

Mark 1.00 out of 1.00

🚩 Flag question

$$P(C, F_1, \dots, F_n) = \prod_i P(F_i | C)$$

Select one:

☒ True☒ FalseThe correct answer is ☒**Question 18**

Correct

Mark 1.00 out of 1.00

🚩 Flag question

$$P(C, F_1, \dots, F_n) = P(C) \times \prod_i P(C | F_i)$$

Select one:

☒ True☒ FalseThe correct answer is ☒**Question 19**

Correct

Mark 1.00 out of 1.00

🚩 Flag question

$$P(C, F_1, \dots, F_n) = P(C) \times \prod_i P(F_i)$$

Select one:

☒ True☒ False

Question 20

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

$$P(C, F_1, \dots, F_n) = P(C) \times \prod_i P(F_i | C, F_1, \dots, F_{i-1})$$

Select one:



The correct answer is

Information

🚩 Flag question

Which of the following statements are true, given this network?

Question 21

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

For all i, j , $P(F_i, F_j | C) = P(F_i | C)P(F_j | C)$.

Select one:



The correct answer is

Question 22

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

If C is not given, the F_i can be dependent.

Select one:



The correct answer is

Question 23

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

All the F_i are pairwise independent.

Select one:



$$P(F_1) \neq P(F_2)$$

Pairwise indep:
 $P(A_1, A_2) = P(A_1)P(A_2)$

Mutually indep if:
 - pairwise indep
 - $P(A_1, A_2, A_3) = P(A_1)P(A_2)P(A_3)$

The correct answer is

Question 24

Correct

Mark 1.00 out of 1.00

🚩 Flag question

The sum over $P(C)$ is 1.

Select one:



The correct answer is

Question 25

Not answered

Not graded

🚩 Flag question

[just for fun; can't give extra points] A question to the experienced ML people among you:
 Do you know the name of this model, if C stands for the class of an object, and the F_i for features?

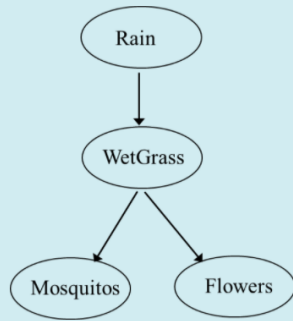
Answer:

The correct answer is:

Information
🚩 Flag question

Inference in Bayesian Networks

Consider the following slightly extended model of the climate in Death Valley and assume that all four variables Rain (R), WetGrass (W), Mosquitos (M) and Flowers (F) are boolean:



Information
🚩 Flag question

Which of the following statements must be true, given the structure of the network?

Question 26
Incorrect
Mark 0.00 out of 1.00
🚩 Flag question

$$P(M) = P(M \mid F)$$

Select one:



The correct answer is



Question 27
Incorrect
Mark 0.00 out of 1.00
🚩 Flag question

$$P(M \mid R) = P(M \mid F, R)$$

Select one:



The correct answer is



Question 28
Correct
Mark 1.00 out of 1.00
🚩 Flag question

$$P(F \mid W) = P(F \mid W, M)$$

Select one:



Question 29

Correct

Mark 1.00 out of 1.00

🚩 Flag question

$$P(W \mid M) = P(W \mid M, F)$$

Select one:



The correct answer is

**Question 30**

Correct

Mark 1.00 out of 1.00

🚩 Flag question

$$P(W \mid F) = P(W \mid F, R)$$

Select one:



The correct answer is

**Question 31**

Correct

Mark 1.00 out of 1.00

🚩 Flag question

$$P(R) = P(R \mid M)$$

Select one:

**Information**

🚩 Flag question

Consider the query $P(R \mid m, \neg f)$
(how probable is it that it rained, given that we see mosquitos, but no flowers).

Question 32

Correct

Mark 2.00 out of 2.00

🚩 Flag question

Would Likelihood Weighting be able to compute a correct ('consistent') estimate for this query?

Select one:



The correct answer is

**Question 33**

Correct

Mark 2.00 out of 2.00

🚩 Flag question

Only for the following question: Assume that, for sampling a value for a variable, we had a defective algorithm that simply samples according to a uniform distribution, i.e., picks each value with the same probability (i.e., 50% in the case of a boolean variable).
If we sampled 100 events from this network, what is the expected number of samples that would be rejected by **Rejection Sampling** (for the above query)?

Answer



We have two binary variables, that means we have 4 possible combinations. Each combination is equally possible as we sample both variables perfectly random with a 50:50 distribution. Therefore each of the 4 combinations is 25% of the whole 100 samples. Only one of the combinations is right, therefore 75% of the combinations have to be rejected.

The correct answer is



Question 34

Correct

Mark 2.00 out of 2.00

Flag question

How many edges (links) would the "Mutilated Network" have that corresponds to this query?

Answer: ✓

The correct answer is

Question 35

Incorrect

Mark 0.00 out of 2.00

Flag question

To answer this query with exact inference, how many different atomic events do we need to sum over?

Answer: ✗

$R', W^0 | R', M'(W^0, F^0) | W^1$
Forward

The correct answer is

Question 36

Incorrect

Mark 0.00 out of 2.00

Flag question

Which of the following expressions correctly describes the Resampling Distribution for variable R in Gibbs Sampling?

Select one:

☐ $\frac{1}{Z} \times P(R | M, F)$

$1/Z * (P(R|mb(R)))$

$mb(R) = W$ (parents, children, and parents of children)

☐ $\frac{1}{Z} \times P(R | W)$

☐ $\frac{1}{Z} \times P(W | R)$

☐ $\frac{1}{Z} \times P(R)$

☐ $\frac{1}{Z} \times P(R)$

☐ $\frac{1}{Z} \times P(M, F | R)$

Your answer is incorrect.

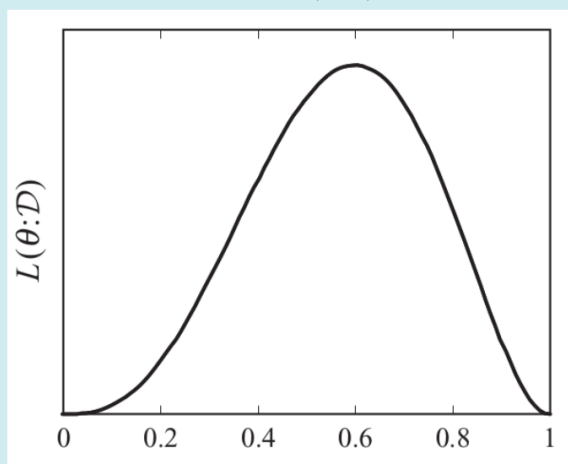
The correct answer is

Information

Flag question

Bayesian Parameter Learning

Consider the example of parameter learning for our thumbtack, from the lecture slides, where we had the following likelihood function, based on $N = 5$ observations, of which $k = 3$ were heads, and $(N - k) = 2$ were tails (which leads to a maximum likelihood estimate $\hat{\theta}_{ML} = 0.6$):



Question 37

Correct

Mark 2.00 out of 2.00

🚩 Flag question

Now assume we perform Bayesian parameter estimation. We believe that thumbtacks behave like fair coins, which we model with a symmetric prior that has its maximum at $\theta = 0.5$.

Will the resulting MAP estimate θ_{MAP} be ...

Select one:

☒ smaller than $\hat{\theta}_{ML}$.

☐ exactly at 0.5.

☐ the same as $\hat{\theta}_{ML}$.

☐ larger than $\hat{\theta}_{ML}$.

Your answer is correct.

The correct answer is: ☒ smaller than $\hat{\theta}_{ML}$.

Question 38

Correct

Mark 2.00 out of 2.00

🚩 Flag question

If we now increase the training data to ten times as many observations ($N = 50$), with the same empirical distribution (30 heads, 20 tails), will the resulting MAP estimate ...

Select one:

☒ a. move closer to $\hat{\theta}_{ML}$.

☐ b. stay the same, but with a higher probability.

☐ c. move farther away from $\hat{\theta}_{ML}$.

Your answer is correct.

The correct answer is: ☒ a. move closer to $\hat{\theta}_{ML}$.

Information

🚩 Flag question

Structure Learning

Indicate which of the following statements is correct.

Question 39

Incorrect

Mark 0.00 out of 2.00

🚩 Flag question

If we replaced the likelihood term in the BIC Score with a constant value, which model would we most likely learn from a dataset (if we did a complete search):

Select one:

☒ a. the fully unconnected model with no parent links

☐ b. a model with $\approx \log M/2$ parameters

☐ c. the fully connected model with all variables connected

Your answer is incorrect.

The correct answer is: ☒ b. a model with $\approx \log M/2$ parameters

Question 40

Correct

Mark 2.00 out of 2.00

🚩 Flag question

If we multiplied the likelihood term in the BIC Score with a constant > 1 , would this

Select one:

☒ a. increase the chance of overfitting.

☐ b. decrease the chance of overfitting.

Your answer is correct.

The correct answer is: ☒ a. increase the chance of overfitting.

Information
Flag question

Hidden Markov Models (HMMs)

Which of the following statements are true?

Question 41

Incorrect
Mark 0.00 out of 1.00
Flag question

If all distributions in the state transition model were uniform (i.e., each state would have the same probability to transition to any other state), it would be impossible to predict future state distributions.

Select one:

☒ True
☒ False

The correct answer is

Question 42

Incorrect
Mark 0.00 out of 1.00
Flag question

The Viterbi algorithm gives a more precise estimate of the state distribution at a time point t than the forward algorithm.

Select one:

☒ True

☐ False

The correct answer is

Question 43

Correct
Mark 1.00 out of 1.00
Flag question

If all distributions in the observation model were uniform (i.e., all observations were equally probable in a given state), the forward algorithm would give the same result with and without the conditioning step.

Select one:

☒ True

The correct answer is

Question 44

Correct
Mark 1.00 out of 1.00
Flag question

The Baum-Welch algorithm constructs a HMM step by step, from left to right.

Select one:

☒ True

The correct answer is

Question 45

Correct
Mark 1.00 out of 1.00
Flag question

The pointwise product of forward and backward message sums to 1.0.

Select one:

☒ True

Question 46

Correct
Mark 1.00 out of 1.00
Flag question

The smoothing algorithm gives smoother state trajectories than the forward algorithm, but less reliable results because it relies on future observations.

Select one:

☒ True

The correct answer is